Order-three unistochastic matrices form a monoid: exact compression of measured qutrit dynamics

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An order-three unistochastic matrix has entries $B(U)_{ij} = |U_{ij}|^2$ for a unitary $U \in U(3)$. Whether the product of two such matrices is always unistochastic is the three-dimensional instance of a previously stated monoid conjecture. We prove that it is. The proof has three ingredients. First, when either factor is orthostochastic, an exterior-algebra construction produces three certificate vectors whose norms are the link lengths in the exact qutrit unistochasticity criterion and whose sum vanishes. Second, the standard phase parametrization of a generic 3×3 unitary makes its squared-modulus matrix an affine chord between two orthostochastic endpoints. Third, the Heron witness for unistochasticity is concave along this chord after multiplication by an arbitrary bistochastic matrix. Endpoint nonnegativity therefore propagates to the entire chord, while compactness includes all singular charts. In the corresponding qutrit experiment, arbitrary unitary controls separated by a complete, unrecorded basis measurement have exactly the same full transition channel as one same-dimension unitary control followed by measurement. The result is exact and finite dimensional; it uses neither an ancillary dimension nor an approximation.

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I. INTRODUCTION

For a unitary matrix $U \in U(d)$, the matrix of transition probabilities

$$B(U)_{ij} = |U_{ij}|^2 \tag{1}$$

is bistochastic. Matrices admitting a representation of the form Eq. (1) are called unistochastic. Their geometry is already nontrivial in order three: they form a proper, nonconvex subset of the Birkhoff polytope, but membership is decided exactly by a triangle-closure condition[1]. Unistochastic matrices also describe the classical transition laws obtained when a finite quantum system is prepared and measured in a fixed basis around a unitary evolution.

Rajchel-Mieldzioć *et al.* introduced bracelet matrices, an outer approximation to the unistochastic set, and conjectured that bracelet matrices form a monoid under ordinary multiplication[2]. They proved closure under multiplication by a factorisable bistochastic matrix. In order three the bracelet and unistochastic sets coincide, so their conjecture specializes to the question

$$A, C \in \mathcal{U}_3 \quad \Longrightarrow \quad CA \in \mathcal{U}_3. \tag{2}$$

The known factorisable-multiplier result does not cover every orthostochastic matrix in order three[3], and the previously proved circulant case is also a strict specialization. Generalized or dilated unistochastic representations[4,5] enlarge the realization space and therefore do not decide the same-size question (2).

This paper proves Eq. (2) in full. The central observation is that the boundary-to-interior geometry of $\mathcal{U}_3$ is compatible with a concave membership witness. An exterior-algebra certificate first establishes closure when one factor is orthostochastic. The usual three-angle, one-phase parametrization of $U(3)$ then expresses a generic unistochastic matrix as a chord between two such endpoint matrices. Along that chord, the order-three Heron witness has a negative-semidefinite quadratic coefficient. This turns endpoint closure into closure for arbitrary factors.

We keep the physical and mathematical statements separate. The theorem is about exact 3×3 matrices. Its physical corollary concerns a fixed qutrit basis, complete dephasing before and after each control, and the entire channel rather than one chosen input distribution. It

does not say that two controls can be generated by one time-independent Hamiltonian, and it does not extend to dimension four.

II. MEASURED QUTRITS AND THE ORDER-THREE CRITERION

Let $\{|0\rangle, |1\rangle, |2\rangle\}$ be a fixed basis of $\mathbb{C}^3$, and define complete dephasing in this basis by

$$\Delta(X) = \sum_{j=0}^{2} |j\rangle\langle j|X|j\rangle\langle j|. \tag{3}$$

For $U \in U(3)$, set

$$\Phi_U = \Delta \circ \mathrm{Ad}_U \circ \Delta, \qquad \mathrm{Ad}_U(X) = UXU^\dagger. \tag{4}$$

We use column probability vectors, so $p' = B(U)p$. If V follows U and the intermediate measurement record is discarded, the transition matrix is $B(V)B(U)$.

Write $\mathcal{B}_3$ for the set of nonnegative 3×3 matrices whose rows and columns each sum to one, $\mathcal{U}_3 = \{B(U) : U \in U(3)\}$, and $\mathcal{O}_3 = \{B(Q) : Q \in O(3)\}$. To state the exact order-three membership test, define for $M \in \mathcal{B}_3$

$$L_i(M) = \sqrt{M_{i0}M_{i1}}, \qquad i = 0, 1, 2, \tag{5}$$

and

$$h(M) = 2 \sum_{0 \le i < j \le 2} L_i^2 L_j^2 - \sum_{i=0}^{2} L_i^4. \tag{6}$$

The qutrit criterion of Ref.[1] can be written as

$$M \in \mathcal{U}_3 \quad \Longleftrightarrow \quad 2 \max_i L_i \le L_0 + L_1 + L_2 \quad \Longleftrightarrow \quad h(M) \ge 0. \tag{7}$$

For completeness, the first equivalence follows by attempting to make the first two columns of a unitary orthogonal. After the first column is chosen real and nonnegative, the condition is

$$\sum_{i=0}^{2} L_i e^{\mathrm{i}\alpha_i} = 0, \tag{8}$$

which is possible exactly when three segments of lengths L_i close. The first two columns are then orthonormal; their one-dimensional orthogonal complement supplies the third column, whose moduli follow from the row sums of M. This argument includes a saturated triangle and zero-length links. The second equivalence follows from the Heron factorization

$$\begin{aligned} h(M) = (L_0 + L_1 + L_2)(-L_0 + L_1 + L_2) \\ \times (L_0 - L_1 + L_2)(L_0 + L_1 - L_2). \end{aligned} \tag{9}$$

We shall also use four coordinates for $\mathcal{B}_3$:

$$M = \begin{pmatrix} a & b & 1-a-b \\ c & d & 1-c-d \\ 1-a-c & 1-b-d & a+b+c+d-1 \end{pmatrix}. \tag{10}$$

Direct substitution in Eq. (6) gives the polynomial identity

$$h(M) = 4abcd - \big(a + b + c + d - 1 - ad - bc\big)^2. \tag{11}$$

No square root or division occurs in Eq. (11), which is useful on the boundary of the polytope.

III. AN EXTERIOR CERTIFICATE AT ORTHOSTOCHASTIC ENDPOINTS

We begin with a multiplier class larger than the previously used factorisable subset.

Lemma 1. *If $A \in \mathcal{U}_3$ and $C \in \mathcal{O}_3$, then $CA \in \mathcal{U}_3$. By transposition, the same conclusion holds if the right factor is orthostochastic.*

Proof. Choose $P \in U(3)$ and $Q \in O(3)$ such that

$$A_{kj} = |P_{kj}|^2, \qquad C_{ik} = Q_{ik}^2. \tag{12}$$

For the first two columns of P, and for every output row i, define $x_i, y_i \in \mathbb{C}^3$ by

$$(x_i)_k = Q_{ik} P_{k0}, \qquad (y_i)_k = Q_{ik} P_{k1}. \tag{13}$$

Use the convention that the Hermitian inner product is antilinear in its first entry, and form

$$z_i = \big(\langle x_i, y_i\rangle,\, x_i \wedge y_i\big) \in \mathbb{C} \oplus \Lambda^2\mathbb{C}^3. \tag{14}$$

The complex Lagrange identity yields

$$\begin{aligned} \|z_i\|^2 &= |\langle x_i, y_i\rangle|^2 + \|x_i \wedge y_i\|^2 \\ &= \|x_i\|^2\|y_i\|^2 = (CA)_{i0}(CA)_{i1}. \end{aligned} \tag{15}$$

Thus $\|z_i\|$ is exactly the link length $L_i(CA)$.

The scalar components sum to zero because the first two columns of P are orthogonal:

$$\sum_i \langle x_i, y_i\rangle = \sum_k \overline{P_{k0}} P_{k1} \sum_i Q_{ik}^2 = 0. \tag{16}$$

For each $k < \ell$, the corresponding exterior components also cancel:

$$\sum_i \big[(x_i)_k (y_i)_\ell - (x_i)_\ell (y_i)_k\big] = (P_{k0}P_{\ell 1} - P_{\ell 0}P_{k1}) \sum_i Q_{ik}Q_{i\ell} = 0. \tag{17}$$

Consequently $z_0 + z_1 + z_2 = 0$. The norm triangle inequality gives

$$L_i(CA) = \|z_i\| \le \sum_{j \ne i} \|z_j\| \quad (i = 0, 1, 2), \tag{18}$$

so Eq. (7) proves $CA \in \mathcal{U}_3$.

The construction can also be read as a direct nonnegative certificate. If $g_{rs} = \langle z_r, z_s\rangle$, then $z_2 = -z_0 - z_1$ and

$$\begin{aligned} h(CA) &= 4\big[g_{00}g_{11} - (\operatorname{Re} g_{01})^2\big] \\ &= 4\big[g_{00}g_{11} - |g_{01}|^2 + (\operatorname{Im} g_{01})^2\big] \ge 0. \end{aligned} \tag{19}$$

The last inequality is the Gram determinant inequality plus a real square. Finally, $(AC)^T = C^T A^T$ and both the unistochastic and orthostochastic sets are invariant under transposition. This proves the reversed-factor statement. □

The proof never assumes positive entries. In particular, zero-link and saturated products are already included and require no limiting phase formula.

IV. THE ORTHOSTOCHASTIC CHORD OF A QUTRIT FACTOR

The next ingredient moves from orthostochastic endpoints to an arbitrary unistochastic factor. A generic 3×3 unitary, modulo row and column rephasings, has the standard three-angle, one-phase form[6]

$$U_{\mathrm{CKM}}(\delta) = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-\mathrm{i}\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{\mathrm{i}\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{\mathrm{i}\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{\mathrm{i}\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{\mathrm{i}\delta} & c_{23}c_{13} \end{pmatrix}, \tag{20}$$

where $s_{jk} = \sin\theta_{jk}$ and $c_{jk} = \cos\theta_{jk}$. The row and column phases do not change squared moduli. The squared-modulus matrix $C(\xi) = B(U_{\mathrm{CKM}})$ depends on the remaining phase only through $\xi = \cos\delta$, and the dependence is affine. Explicitly,

$$\frac{dC}{d\xi} = \kappa \begin{pmatrix} 0 & 0 & 0 \\ 1 & -1 & 0 \\ -1 & 1 & 0 \end{pmatrix}, \qquad \kappa = 2s_{12}c_{12}s_{13}s_{23}c_{23}. \tag{21}$$

It follows that

$$C(\xi) = \frac{1+\xi}{2}C(+1) + \frac{1-\xi}{2}C(-1), \qquad -1 \le \xi \le 1. \tag{22}$$

At $\xi = \pm 1$ one may take $\delta = 0$ or π; the representative Eq. (20) is then real and unitary, hence orthogonal. Thus every generic member of $\mathcal{U}_3$ lies on an affine chord with endpoints in $\mathcal{O}_3$.

For later use, it is enough that this construction holds on the generic dense chart. Indeed, generic unitaries are dense in $U(3)$, and the map $U \mapsto B(U)$ is continuous. Moreover, $\mathcal{U}_3$ is compact, because it is the continuous image of compact $U(3)$, and hence it is closed. These facts will include collapsed chords and all other singular charts without dividing by a vanishing mixing angle.

V. CONCAVITY AND THE MONOID THEOREM

Fix an arbitrary $A \in \mathcal{B}_3$ and let $C(t)$ move affinely in the direction Eq. (21). For $M(t) = C(t)A$, the first row is constant, while the second and third rows move oppositely. Therefore the four coordinates of Eq. (10) can be written as

$$a(t) = a, \qquad b(t) = b, \qquad c(t) = c + tr, \qquad d(t) = d + ts. \tag{23}$$

Because $M(t)$ is bistochastic throughout the physical chord, $a \geq 0$, $b \geq 0$, and $a + b \leq 1$. Substitution of Eq. (23) in Eq. (11) shows that the coefficient of t^2 in $h(M(t))$ is

$$\begin{aligned} K &= 4abrs - \big[(1-b)r + (1-a)s\big]^2 \\ &= \begin{pmatrix} r & s \end{pmatrix} \begin{pmatrix} -(1-b)^2 & ab + a + b - 1 \\ ab + a + b - 1 & -(1-a)^2 \end{pmatrix} \begin{pmatrix} r \\ s \end{pmatrix}. \end{aligned} \tag{24}$$

The two diagonal entries of the displayed symmetric matrix are nonpositive, and its determinant is

$$4ab(1 - a - b) \geq 0. \tag{25}$$

The matrix is therefore negative semidefinite, so $K \leq 0$. We have proved the following statement, including boundary values of a and b.

Lemma 2. *For every $A \in \mathcal{B}_3$, the function $\xi \mapsto h(C(\xi)A)$ is concave on every qutrit CKM chord.*

We can now state the main result.

Theorem 1. *The order-three unistochastic set is a monoid under ordinary matrix multiplication. Equivalently, for every $A, C \in \mathcal{U}_3$, one has $CA \in \mathcal{U}_3$.*

Proof. First suppose that C is represented in a generic CKM chart, and write it as $C(\xi)$ in Eq. (22). Lemma 1 gives

$$h(C(-1)A) \geq 0, \qquad h(C(+1)A) \geq 0, \tag{26}$$

because both endpoint matrices are orthostochastic. Lemma 2 and concavity then imply

$$h(C(\xi)A) \geq \frac{1-\xi}{2} h(C(-1)A) + \frac{1+\xi}{2} h(C(+1)A) \geq 0. \tag{27}$$

The product $C(\xi)A$ is bistochastic, so the exact criterion Eq. (7) makes it unistochastic.

For arbitrary $C = B(V) \in \mathcal{U}_3$, choose generic unitaries $V_n \to V$. The preceding argument gives $B(V_n)A \in \mathcal{U}_3$ for every n, and $B(V_n)A \to CA$. Closedness of $\mathcal{U}_3$ gives $CA \in \mathcal{U}_3$. The identity matrix is unistochastic, completing the monoid statement. □

The proof is specific to order three in two independent ways. The CKM phase gives a one-dimensional affine chord, and the three-link condition is both necessary and sufficient for unistochasticity. In order four, bracelet conditions cease to be sufficient and the unistochastic set is not closed under arbitrary products[1,2].

VI. CONSTRUCTING THE COMPRESSED CONTROL

The theorem is existential at the level of A and C, but the order-three criterion gives a direct construction of a unitary realizing the product. Given $M = CA$, compute the three lengths in Eq. (5). Because $h(M) \geq 0$, choose phases α_i that close the triangle in Eq. (8). Define the first two columns of W by

$$W_{i0} = \sqrt{M_{i0}}, \qquad W_{i1} = \sqrt{M_{i1}}e^{\mathrm{i}\alpha_i}. \tag{28}$$

Column normalization follows from bistochasticity and their inner product vanishes by construction. Choose the third column as either unit vector in their one-dimensional orthogonal complement. The resulting square matrix is unitary. Its row normalization gives

$$|W_{i2}|^2 = 1 - |W_{i0}|^2 - |W_{i1}|^2 = 1 - M_{i0} - M_{i1} = M_{i2}, \tag{29}$$

and hence $B(W) = M$. Saturated and zero-link cases are handled by choosing collinear or arbitrary zero-link phases before taking the orthogonal complement; no cosine-law division is required.

We now translate this construction back to the measured qutrit. From Eqs. (3)–(4), for every operator X,

$$\Phi_U(X) = \sum_{i,j=0}^{2} B(U)_{ij} \langle j|X|j\rangle \, |i\rangle\langle i|. \tag{30}$$

Therefore

$$\Phi_V \circ \Phi_U = \Phi_W \quad \text{whenever} \quad B(W) = B(V)B(U). \tag{31}$$

This is equality of linear maps, not agreement on one input probability vector. It consequently remains true after tensoring both sides with the identity channel on any finite-dimensional reference system. The intermediate record is discarded, and W acts on the same three-level system. The conclusion does not assert $W = VU$, does not preserve an intermediate outcome, and does not impose a common Hamiltonian or a control energy bound.

Associativity immediately allows the matrix statement to be iterated over any finite product in $\mathcal{U}_3$. Operationally, each iteration still assumes the same fixed basis and complete dephasing at the specified interfaces.

VII. EXACT CHECKS AND USE OF GENERATIVE AI

The proof above is analytic and does not rely on a finite search. Two accompanying SymPy programs provide reproducible algebraic checks. The first verifies the complex Lagrange identity, all cancellation contractions, and Eq. (19), including strict and zero-entry examples. The second expands Eqs. (11), (24), and (25), verifies the CKM endpoints symbolically, and completes an independent exact interior example to a unitary W. The environment and commands are specified in the accompanying reproducibility file. Numerical sampling used during exploration is not evidence for the universal theorem.

OpenAI Codex (a GPT-5-based coding agent, OpenAI; accessed September 2026) was used in literature triage, proof exploration, symbolic-checking code, referee-style computational rechecks, and manuscript drafting. Its role was to accelerate algebraic exploration and to expose the argument to distinct symbolic and numerical checks. The named author must critically review the scientific content, citations, code, and prose before submission and remains responsible for their integrity, accuracy, and originality; this local draft does not represent that human review as completed.

VIII. CONCLUSION

Order-three unistochastic matrices are closed under ordinary multiplication. The proof combines an exterior-algebra endpoint certificate with the affine phase geometry of a qutrit unitary and a negative-semidefinite concavity form. It resolves the order-three specialization

of the bracelet-monoid conjecture without restricting either factor to the previously known factorisable or circulant classes.

For a fixed qutrit basis, the result gives exact compression of two arbitrary unitary-and-dephasing steps into one unitary-and-dephasing step at the level of the full channel. This is a finite-dimensional structural statement. It does not supply a higher-dimensional closure theorem, an implementation by a single time-independent Hamiltonian, or an experimentally costed control protocol. Those distinctions are essential when using the monoid property as a statement about physical dynamics.

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DATA AVAILABILITY

No experimental data were generated. Exact symbolic verification scripts, their reference outputs, and environment instructions accompany this draft. They audit algebraic identities and examples; the universal conclusion rests on the analytic proof in Secs. III–V.

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